

*ident, x*  
*X*  
*U*

|           |                                                                                                                                                                                                                                                     |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $exp, e$  | $::=$<br>$ $ $x$<br>$ $ $U.$ <b>Term</b><br>$ $ $[e]_c^h$<br>$ $ $()$<br>$ $ $(exp, exp')$<br>$ $ <b>function</b> $x : type \rightarrow exp$ bind $x$ in $exp$<br>$ $ $exp\ exp'$<br>$ $ $\{exp/x\}exp'$ M<br>$ $ $(exp)$ M<br>$ $ <b>Foo</b> $val$ |
| $val, v$  | $::=$<br>$ $ $[v]_c^h$<br>$ $ $()$<br>$ $ $(val, val')$<br>$ $ <b>function</b> $x : t \rightarrow exp$                                                                                                                                              |
| $M$       | $::=$<br>$ $ $[t, v]$                                                                                                                                                                                                                               |
| $S$       | $::=$<br>$ $ $[X : K, t]$ bind $X$ in $t$                                                                                                                                                                                                           |
| $Sab$     | $::=$<br>$ $ $[X : Kab, t]$ bind $X$ in $t$                                                                                                                                                                                                         |
| $K$       | $::=$<br>$ $ <b>Type</b><br>$ $ <b>Eq</b> $(t)$                                                                                                                                                                                                     |
| $Kab$     | $::=$<br>$ $ <b>Type</b>                                                                                                                                                                                                                            |
| $m$       | $::=$<br>$ $ $e$<br>$ $ <b>let</b> $U = M : S \in m$ bind $U$ in $m$                                                                                                                                                                                |
| $h$       | $::=$<br>$ $ <b>hash</b> $(M : Sab)$                                                                                                                                                                                                                |
| $c$       | $::=$<br>$ $ <b>empty</b><br>$ $ $c$ <b>union</b> $c'$<br>$ $ $c$ <b>intersect</b> $c'$ M<br>$ $ $\{h\}$                                                                                                                                            |
| $type, t$ | $::=$<br>$ $ $X$                                                                                                                                                                                                                                    |

|                |       |                          |   |
|----------------|-------|--------------------------|---|
|                |       | $U.\mathbf{Type}$        |   |
|                |       | $\mathbf{unit}$          |   |
|                |       | $type * type'$           |   |
|                |       | $type \rightarrow type'$ |   |
| $E$            | $::=$ |                          |   |
|                |       | $\mathbf{empty}$         |   |
|                |       | $E, x : t$               |   |
|                |       | $E, x : t, E'$           |   |
| $formula$      | $::=$ |                          |   |
|                |       | $judgement$              |   |
|                |       | $\mathbf{not} (formula)$ |   |
|                |       | $x \in \mathbf{dom} (E)$ | M |
|                |       | $x \mathbf{notin} c$     | M |
|                |       | $h \mathbf{notin} c$     | M |
| $terminals$    | $::=$ |                          |   |
|                |       | $\lambda$                |   |
|                |       | $\longrightarrow$        |   |
|                |       | $\rightarrow$            |   |
|                |       | $\vdash$                 |   |
|                |       | $\in$                    |   |
| $Jtype$        | $::=$ |                          |   |
|                |       | $E \vdash \mathbf{ok}$   | M |
|                |       | $E \vdash e : t$         |   |
|                |       | $v : c$                  |   |
| $Jop$          | $::=$ |                          |   |
|                |       | $e \longrightarrow e'$   |   |
|                |       | $e \longrightarrow_c e'$ |   |
| $judgement$    | $::=$ |                          |   |
|                |       | $Jtype$                  |   |
|                |       | $Jop$                    |   |
| $user\_syntax$ | $::=$ |                          |   |
|                |       | $ident$                  |   |
|                |       | $X$                      |   |
|                |       | $U$                      |   |
|                |       | $exp$                    |   |
|                |       | $val$                    |   |
|                |       | $M$                      |   |
|                |       | $S$                      |   |
|                |       | $Sab$                    |   |
|                |       | $K$                      |   |
|                |       | $Kab$                    |   |
|                |       | $m$                      |   |

$|$   $h$   
 $|$   $c$   
 $|$   $type$   
 $|$   $E$   
 $|$   $formula$   
 $|$   $terminals$

$E \vdash \mathbf{ok}$

$$\begin{array}{c}
\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \text{EOK\_EMPTY} \\
\frac{E \vdash \mathbf{ok} \quad \mathbf{not}(x \in \mathbf{dom}(E))}{E, x : t \vdash \mathbf{ok}} \text{EOK\_IDENT}
\end{array}$$

$E \vdash e : t$

$$\begin{array}{c}
\frac{E, x : t, E' \vdash \mathbf{ok}}{E, x : t, E' \vdash x : t} \text{EET\_IDENT} \\
\frac{E \vdash \mathbf{ok}}{E \vdash () : \mathbf{unit}} \text{EET\_UNIT} \\
\frac{E \vdash e_1 : t_1 \quad E \vdash e_2 : t_2}{E \vdash (e_1, e_2) : t_1 * t_2} \text{EET\_PAIR} \\
\frac{E, x : t_1 \vdash e : t_2}{E \vdash \mathbf{function } x : t_1 \rightarrow e : t_1 \rightarrow t_2} \text{EET\_FUN} \\
\frac{E \vdash e : t_1 \rightarrow t_2 \quad E \vdash e' : t_1}{E \vdash e e' : t_2} \text{EET\_APP}
\end{array}$$

$v : c$

$$\begin{array}{c}
\frac{v : c' \quad h \mathbf{notin } c' \mathbf{intersect } c}{[v]_{c'}^h : c} \text{WFFVAL\_COLOURED} \\
\frac{}{() : c} \text{WFFVAL\_UNIT} \\
\frac{v : c \quad v' : c}{(v, v') : c} \text{WFFVAL\_PAIR} \\
\frac{}{\mathbf{function } x : t \rightarrow exp : c} \text{WFFVAL\_FUN}
\end{array}$$

$e \longrightarrow e'$

$$\begin{array}{c}
\frac{}{(\mathbf{function } x : t \rightarrow e) v \longrightarrow \{v/x\}e} \text{JO\_RED\_FUN} \\
\frac{e_1 \longrightarrow e'_1}{e_1 e_2 \longrightarrow e'_1 e_2} \text{JO\_RED\_CONTEXT1}
\end{array}$$

$$\frac{e_2 \longrightarrow e'_2}{v_1 \ e_2 \longrightarrow v_1 \ e'_2} \quad \text{JO\_RED\_CONTEXT2}$$

$$\frac{e_1 \longrightarrow e'_1}{(e_1, e_2) \longrightarrow (e'_1, e_2)} \quad \text{JO\_RED\_CONTEXT3}$$

$$\frac{e_2 \longrightarrow e'_2}{(v_1, e_2) \longrightarrow (v_1, e'_2)} \quad \text{JO\_RED\_CONTEXT4}$$

$$\boxed{e \longrightarrow_c e'}$$

$$\frac{e \longrightarrow_{c'} e'}{[e]_{c'}^h \longrightarrow_c [e']_{c'}^h} \quad \text{JO\_CRED\_COLOURED}$$

Definition rules: 17 good 0 bad

Definition rule clauses: 35 good 0 bad